

NETWORK INFRASTRUCTURE · PROJECT DOCUMENTATION

Building Management System (BMS) Network Infrastructure

Design and Implementation of a Segmented Enterprise
Network using VLANs, Trunking, and Inter-VLAN Routing

Platform: Cisco Packet Tracer

Core Device: Cisco 3560 Multilayer Switch

Scope: VLAN Segmentation · 802.1Q Trunking · Inter-VLAN Routing · Network Troubleshooting

1. Introduction

Project Name

Building Management System (BMS) Network Infrastructure Design and Implementation

Project Objective

To design and implement a secure enterprise network for a Building Management System using VLAN segmentation, trunking, management VLANs, and Inter-VLAN Routing.

The project simulates a real-world commercial building where multiple departments require network isolation while accessing centralized building services.

2. Business Scenario

A commercial building contains multiple departments:

Department	VLAN
P2 Operations Team	VLAN 4
P3 Maintenance Team	VLAN 13
P4 Security Team	VLAN 7
P5 BMS Infrastructure	VLAN 8
Network Management	VLAN 100

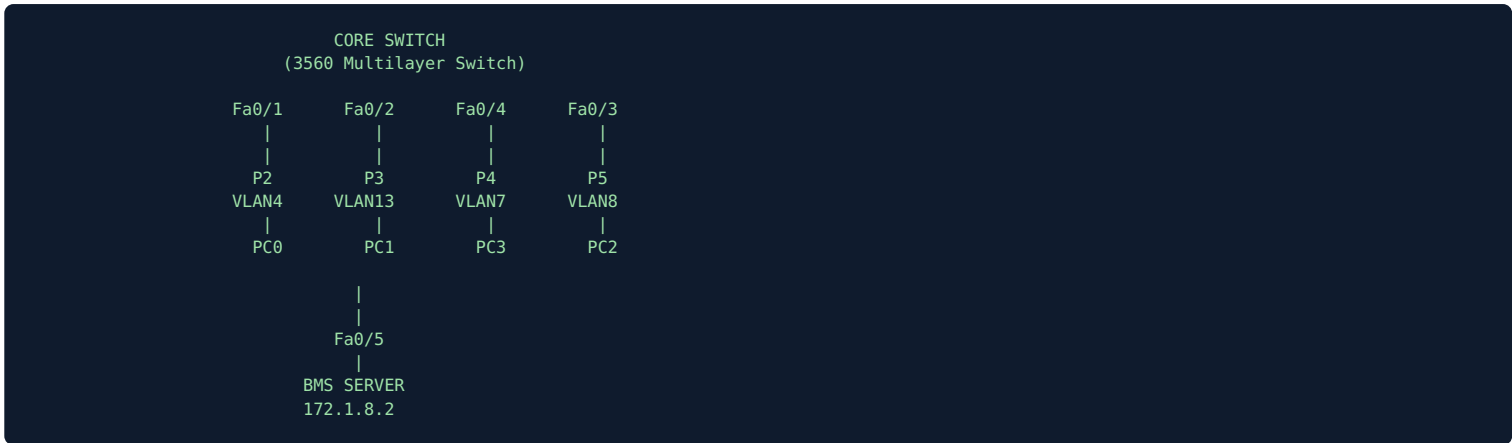
Requirements

- Departments must be isolated.
- All users should access the BMS Server.
- Network devices should have management IPs.
- Routing should be centralized.
- Traffic should flow through a Core Layer-3 Switch.

3. Network Topology

Insert topology screenshot.

Topology Diagram



4. IP Addressing Scheme

VLAN Gateways

VLAN	Gateway
VLAN 4	192.168.4.1
VLAN 13	192.168.13.1
VLAN 7	192.168.7.1
VLAN 8	172.1.8.1
VLAN 100	192.168.14.1

Management IPs

Device	Management IP
Core	192.168.14.1
P2	192.168.14.2
P3	192.168.14.3
P4	192.168.14.4
P5	192.168.14.5

End Devices

Device	IP
PC0	192.168.4.2
PC1	192.168.13.2
PC3	192.168.7.2
PC2	172.1.8.3
Server	172.1.8.2

5. Implementation Process

Step 1: Build Physical Topology

TASKS PERFORMED

- Added one 3560 Multilayer Switch
- Added four 2960 switches
- Added four PCs
- Added one Server
- Connected all devices using copper straight-through cables

VERIFICATION

```
show cdp neighbors
```

Result: Core switch successfully discovered all access switches.

Step 2: VLAN Creation

CREATED VLANS

```
vlan 4
name P2

vlan 13
name P3

vlan 7
name P4

vlan 8
name P5

vlan 100
name MANAGEMENT
```

VERIFICATION

```
show vlan brief
```

Result: All VLANs successfully created.

Step 3: Configure SVIs

CONFIGURED GATEWAY INTERFACES

```
interface vlan 4
ip address 192.168.4.1 255.255.255.0

interface vlan 13
ip address 192.168.13.1 255.255.255.0

interface vlan 7
ip address 192.168.7.1 255.255.255.0

interface vlan 8
ip address 172.1.8.1 255.255.255.0

interface vlan 100
ip address 192.168.14.1 255.255.255.0
```

ENABLED ROUTING

```
ip routing
```

VERIFICATION

```
show ip interface brief
```

Step 4: Configure Trunk Links

CONFIGURED

```
switchport trunk encapsulation dot1q
switchport mode trunk
```

ALLOWED VLANS

Link	VLANs
Core-P2	4, 100
Core-P3	13, 100
Core-P4	7, 100
Core-P5	8, 100

VERIFICATION

```
show interfaces trunk
```

Step 5: Configure Access Switches

CONFIGURED

- Access Ports
- Trunk Ports
- VLAN Interfaces
- Default Gateways

VERIFICATION

```
show vlan brief
show interfaces trunk
```

Step 6: Configure End Devices

Assigned IP addresses manually.

VERIFICATION

```
ping gateway
```

Result: All devices successfully reached their gateways.

6. Troubleshooting Performed

Issue 1

PROBLEM

Could not configure trunk on Core switch.

ERROR

```
Command rejected:
An interface whose trunk encapsulation is Auto
cannot be configured to trunk mode
```

ANALYSIS

Interface encapsulation was set to Auto.

RESOLUTION

```
switchport trunk encapsulation dot1q
switchport mode trunk
```

Result: Trunk link established successfully.

Issue 2

PROBLEM

Server was unreachable from all PCs.

SYMPTOMS

```
ping 172.1.8.2
```

failed from all VLANs.

INVESTIGATION

```
show vlan brief
```

Observed:

```
Fa0/5 present in VLAN 1
```

instead of VLAN 8.

ROOT CAUSE

Server-facing port remained in default VLAN.

RESOLUTION

```
interface fa0/5
switchport mode access
switchport access vlan 8
```

VERIFICATION

```
show interfaces fa0/5 switchport
```

Output:

```
Access Mode VLAN: 8
```

Result: Server became reachable from all VLANs.

Issue 3

PROBLEM

Needed to identify switch-to-switch connections.

INVESTIGATION

```
show cdp neighbors
```

RESULT

Successfully identified:

```
Fa0/1 -> P2
Fa0/2 -> P3
Fa0/3 -> P5
Fa0/4 -> P4
```

7. Verification Tests

VLAN Connectivity

PC0

```
ping 192.168.4.1
```

Success.

PC1

```
ping 192.168.13.1
```

Success.

Inter-VLAN Routing

PC1 to PC0

```
ping 192.168.4.2
```

Success.

PC0 to PC3

```
ping 192.168.7.2
```

Success.

PC0 to PC2

ping 172.1.8.3

Success.

Server Connectivity

Core to Server

ping 172.1.8.2

RESULT

!!!!!
Success rate 100%

All PCs to Server

Successful.

8. Skills Demonstrated

- VLAN Configuration
- Inter-VLAN Routing
- Layer 3 Switching
- Cisco Switching
- IEEE 802.1Q Trunking
- IP Addressing
- Network Troubleshooting
- CDP Verification
- Enterprise Network Design
- Packet Tracer Implementation

9. Conclusion

Successfully designed and implemented a Building Management System network infrastructure. The project achieved departmental segregation using VLANs, centralized routing using a Layer-3 Core Switch, secure management access through VLAN 100, and centralized application access through the BMS Server.

The project demonstrates practical enterprise networking skills including VLAN implementation, trunking, routing, troubleshooting, and end-to-end connectivity validation.